

ARCHITECTURAL BLUEPRINT & TECHNICAL SPECIFICATION

Aegis Solana Wallet Risk Scanner

Wallet & Token Auto-Detection and Premium UI/UX Implementation Plan

This blueprint structures the technical roadmap for real Solana RPC queries, deterministic address filters, a state-of-the-art DApp visuals overhaul (neon gauges, scanner hologram, glassmorphism UI/UX), and complete test specifications.

This plan outlines the implementation of wallet/token type filtering, real Solana address analysis (with RPC queries and local heuristics fallback), and a complete UI/UX overhaul to give the application a premium, DApp-like aesthetic.

User Review Required

IMPORTANT BLUEPRINT NOTE

Key Enhancements Proposed:

- Address Type Auto-Detection: We will query the Solana mainnet-beta RPC to determine if an address is a Wallet or a Token (Mint).
- Interactive Filters & Tabs:
- Input/History Filters: A sidebar or toggle list to view past scans, filterable by All, Wallets, or Tokens.
- Result Sub-tabs: We will divide the results into structured tabs: Security Audit, Holdings / Contract Info, and Recent Activity.

Proposed Changes

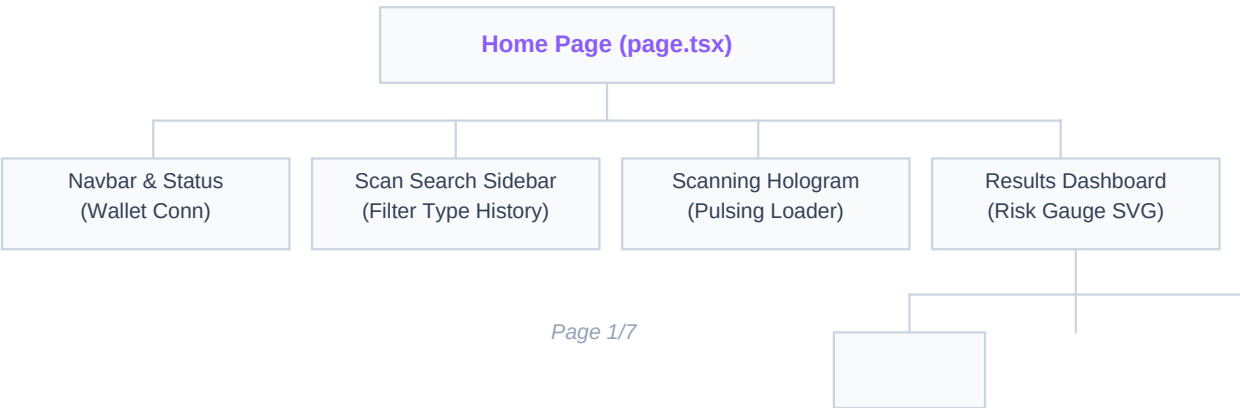
For the Search/Filter For Wallets, the user can search and filter the token holdings list by Risk Category (Safe / Warning / Danger) or Balance.

- Premium Visuals:

We will restructure the application to support robust state management, dedicated UI components, and rich data visualization.

- Full suite of neon accents matching risk statuses: Emerald Green (Safe), Amber Gold (Caution), Rose Red (High Risk), and Electric Violet (Brand/Primary).
- A high-tech "scanning hologram" micro-animation when analysis is running.
- Glowing interactive gauge for the risk score.

SYSTEM ARCHITECTURE & VISUAL COMPONENT MAP



Security
Audit



Holdings
Filter



Activity
Timeline

Backend (API Layer)

MODIFY

route.ts

- Add connection to Solana mainnet-beta (<https://api.mainnet-beta.solana.com>).
- Query `connection.getAccountInfo(pubkey)` to identify owner and data length.
- Standardize the response contract:
- `type: "wallet" | "token"`
- `score: 0-100`
- `details`: Account metadata (e.g. balance, token name, decimals, supply, owner program)
- `analysis`: Customized security checklist items (Rug Risk, Mixer interactions, Authority lock, etc.)
- `holdings`: (For wallets) List of standard mock/real token holdings with search and risk data.
- `transactions`: Color-coded activity log.
- Handle rate-limiting or RPC failures gracefully by using deterministic pseudo-random fallback matching the user's input type.

Components (Frontend Redesign)

MODIFY

globals.css

- Implement custom CSS styles for ambient glow shadows, neon variables, glassmorphism, and keyframe animations (scanner line sweep, holographic pulse).

MODIFY

layout.tsx

- Set up a premium title ("Aegis Solana - Advanced Risk Scanner"), meta description, and import the modern "Outfit" or "Inter" Google Font for a clean cryptographic aesthetic.

MODIFY

InputSection.tsx

- Re-style to a futuristic floating input with an electric violet border glow.
- Add quick-select pills ("Auto-Detect", "Wallet", "Token") to let the user specify or test specific scanner modes.

MODIFY

RiskScore.tsx

- Upgrade the gauge to a double-layered gradient SVG circle with a soft ambient glow.
- Add a counting-up number animation for the score value.

MODIFY

RiskAnalysis.tsx

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Redesign the checklist as interactive, expandable rows with beautiful colored status indicators, description tooltips, and security badges.

NEW

HoldingsList.tsx

- A component specifically for rendering wallet token assets.
- Includes a search bar to filter tokens by name or symbol.
- Includes tabs or pills to filter tokens by risk levels (e.g., "All", "High Risk / Warning", "Safe / Verified").
- Shows a beautifully formatted list of holdings with dollar-values, token symbols, balances, and security indicators.

NEW

TransactionTimeline.tsx

- Shows a list of recent activity (incoming/outgoing transactions, contract interactions, swaps) with type badges and timestamps.

NEW

HistoryPanel.tsx

- An expandable side panel or sidebar listing recent scans.
- Features tabs to filter the history by scan type: All, Wallets, or Tokens.
- Allows quick deletion or reloading of past scans.

MODIFY

page.tsx

- Redesign the main application container to support the dashboard layout (sidebar + main view).
- Implement interactive results tabs: Security Audit, Holdings / Contract Info, and Recent Activity.
- Embed the futuristic pulsing scan loader.

Verification Plan

Automated / Browser Validation

- Open the application locally and check for typescript compile errors.
- Validate that the dev server correctly parses the new CSS variables and Tailwind extensions.

Manual Verification

1. Wallet Scan Test:

- Scan a valid Solana wallet address (e.g. Hw21oR1R21oH7... or a generic valid pubkey).
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Verify that the app detects it as a wallet.

- Test search and filtering inside the "Holdings" list by typing "USDC" or clicking risk filter pills.

2. Token Scan Test:

- Scan a token address.
- Verify that the app detects it as a token contract and displays contract audit specs rather than token holdings.

3. History Filter Test:

- Perform multiple wallet and token scans.
- Go to the history sidebar and filter using "Wallets" and "Tokens" tabs. Verify only matching scans are shown.

4. Visual Quality Review:

- Verify the dark-mode aesthetic, glowing gradients, hover-micro-animations, and responsive layouts.